Problem Set 1 — Solutions

Problem 1: Spaces and Norms

Part 1

2. Newtonian forces acting on an object

- Newtonian forces can be represented as belonging to $\mathbb{R}^2$ for 2D objects or $\mathbb{R}^3$ for 3D objects.
- A value carrying both magnitude and direction, drawn from a reference point — e.g. a force of $[3, 4]$ on an object in the 2D plane, drawn from the origin.

4. Angle of a robot's elbow joint Represented by $\mathbb{R}$ mod 360 is *not* a vector space. It fails on the scalar-distribution axiom, e.g. take $a = \frac{1}{2}$, $u = 200$, $v = 300$:

$$a(u + v) = \tfrac{1}{2} \cdot (500 \bmod 360) = \tfrac{1}{2} \cdot 140 = 70,$$
$$au + av = \tfrac{1}{2} \cdot 200 + \tfrac{1}{2} \cdot 300 = 100 + 150 = 250.$$

So $a(u + v) \neq au + av$, and the scalar-distribution axiom fails.

Acknowledgements

This document was prepared with the aid of the following reference:

- O. Knill, *Linear Algebra and Multivariable Calculus* (Harvard Math 22b, 2019), Lecture 25: `https://people.math.harvard.edu/~knill/teaching/math22b2019/handouts/lecture25.pdf`